

Key Points:

- Three-dimensional temperature and salinity anomaly fields are estimated exclusively from satellite data through a deep learning approach
- Estimated data are at 26 depths (0–2,000 m) and horizontal resolution has been improved to 1/4°, higher than Argo gridded data
- Our model is accurate and reliable for a wide range of research area and period; dynamic fields from the estimated data are also analyzed

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Meng, L., Yan, C., Zhuang, W., Zhang, W., & Yan, X.-H. (2021). Reconstruction of three-dimensional temperature and salinity fields from satellite observations. *Journal of Geophysical Research: Oceans*, 126, e2021JC017605. <https://doi.org/10.1029/2021JC017605>

Received 24 MAY 2021

Accepted 4 NOV 2021

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Reconstruction of Three-Dimensional Temperature and Salinity Fields From Satellite Observations

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Abstract Observation of the ocean is crucial to the studies of ocean dynamics, climate change, and biogeochemical cycle. However, current oceanic observations are patently insufficient, because the in situ observations are of difficulty and high cost while the satellite remote-sensed measurements are mainly the sea surface data. To make up for the shortage of ocean interior data and make full use of the abundant satellite data, here we develop a data-driven deep learning model to estimate ocean subsurface and interior variables from satellite-observed sea surface data. Exclusively and simply using satellite data, three-dimensional ocean temperature and salinity fields are successfully reconstructed, which are at 26 level depths from 0 to 2,000 m. We further design a scheme to increase the horizontal resolution from 1° to 1/4°, which is higher than the Argo gridded data. Estimations from our model are accurate, reliable, and stable for a wide range of research areas and periods. Dynamic height fields that are derived from the estimated temperature and salinity, as well as the associated ocean geostrophic flows, are also calculated and analyzed, which indicates the potentials of our model for reconstructing the ocean circulation fields as well. This study enriches oceanic observations with respect to vertical dimension and horizontal resolution, which can largely make up for the paucity of the subsurface and deep ocean observation, both before and during Argo era. This work also provides some new foundations for and insights into geoscience and climate change fields.

Plain Language Summary Ocean data are important for ocean science and climate change studies; however, oceanic observations are difficult and costly and thus are still very sparse in space and time. Satellites have been providing plentiful oceanic observations, but these data are only at the sea surface. To fully utilize the copious satellite data and to make up for the shortage of ocean interior data. Here, we establish a deep learning model to connect the surface ocean with the subsurface and deep oceans, through which, subsurface and deep ocean temperature and salinity data are estimated from the surface data observed by satellites. We also design a new scheme to improve the horizontal resolution of the obtained data. The results show that our model successfully reconstructed the three-dimensional field data of temperature and salinity. Our model could facilitate ocean science studies by largely enriching the ocean data availability.

1. Introduction

Observations provide us the basic information about the oceans. Nowadays, oceanic observations are collected using many different platforms and instruments. In-site observations provide us the first-hand and genuine data of the ocean; however, the measured data remain temporally and spatially sparse due to the difficulty and costliness of in-site measurements. Traditional methods such as Expendable Bathythermograph, buoy, boat measurements, etc. are short-living, sparse, and mostly nearshore (Stewart, 2008; Talley et al., 2011d); the implementation of the Argo program since 2,004 has substantially enhanced global oceans observations, however, improvements of Argo program are still needed, such as observing density, distribution, and time-span (Argo, 2020; Roemmich et al., 2009, 2019).

Relative to the above, satellite-derived ocean data are copious, continuous, and at large spatial coverages. Satellites have been applied to monitor the ocean since the 1970s, continuously providing products of various ocean parameters (Klemas & Yan, 2014; Talley et al., 2011d), such as sea surface temperature (SST) from microwave radiometers and thermal infrared sensors, sea level from altimeter satellites, sea surface wind

(SSW), sea surface roughness examined by Synthetic Aperture Radar, and sea surface salinity (SSS) from the Aquarius/SAC-D mission and the NASA Soil Moisture Active Passive mission. However, satellite products are constrained at sea surface because the electromagnetic wave of satellite sensors cannot penetrate deep into the seawater (Klemas & Yan, 2014; Talley et al., 2011d).

Nonetheless, subsurface and deep ocean data are of great importance because many important processes and phenomena are located there, such as oceanic eddies, subsurface flows, internal waves (Chen & Tung, 2014; Klemas & Yan, 2014; Meng et al., 2020; Yan et al., 2016). Moreover, oceans function as the main heat sink in the Earth system (Trenberth & Fasullo, 2010, 2013), the heat redistributions within the ocean interior have substantial influences on global mean surface temperature and climate change (Kosaka & Xie, 2013; Trenberth, 2015; Trenberth & Fasullo, 2013; Xie & Kosaka, 2017; Yan et al., 2016). Some recent discussions on the changes and mechanisms of the Atlantic meridional overturning circulation, part of the thermohaline circulation, also highlighted the importance of the observations of subsurface and deep ocean (Bryden et al., 2005; Cunningham et al., 2007; Hu & Fedorov, 2019; Lozier et al., 2019). To this end, retrieving the ocean interior from satellite-observed surface information can greatly enhance data usage and facilitate the studies of three-dimensional (3D) ocean dynamics.

The ocean interior parameters are dynamically connected to the surface ones because the forces at the surface are ascribed to the main changes in the whole ocean (R. X. Huang, 2009; Munk, 1950); and many subsurface processes and phenomena yield surface manifestation, such as internal waves, mixed layer depth, and eddies (Yan et al., 1991, 2006; Yan & Okubo, 1992). These surface-subsurface connections enable the retrieval of subsurface and deep ocean data from the surface information (Ali et al., 2004; Fu & Davidson, 1995; Wu et al., 2012). Generally, the retrieval of ocean interior parameters from surface information is made either dynamically or statistically (or both). The typical dynamical approaches are dynamic-based numerical modeling (simulation and assimilation), as well as models based on direct physical relationship; however, since the direct physical relationship between surface and subsurface data are not fully understood, those models work effectively only under certain conditions or with some certain assumptions (Guinehut et al., 2012; L. Liu et al., 2014; L. Liu et al., 2019; Wang et al., 2013; Yan et al., 2006; Yan & Okubo, 1992). By contrast, statistical approaches are relatively more practical and flexible, which can be implemented in more general situations (Akbari et al., 2017; Holloway & Mengersen, 2018; Klemas & Yan, 2014; Lary et al., 2016).

Regarding the statistical approaches, they can generally be divided into traditional statistical methods, empirical methods, machine learning methods, and deep learning methods. The traditional statistical methods and empirical methods, such as multi-variable regression, are heuristic approaches and some special criteria or prescriptions are normally needed. By contrast, machine learning models can extract features and relations automatically and directly from data and thus achieve remarkable performance currently (Ali et al., 2004; Chu et al., 2000; Fischer, 2000). Until lately, various machine learning methods have been adopted for estimating ocean interior structures, such as artificial neural networks, self-organizing machine, support vector machine, random forest (Akbari et al., 2017; Ali et al., 2004; Su et al., 2015, 2018; Wu et al., 2012). Furthermore, deep learning networks are machine learning networks with deeper hidden layers and larger architecture and thus have larger capacities and can capture more complicated features (Bolton & Zanna, 2019; Goodfellow et al., 2016; Ham et al., 2019). As a specific deep learning algorithm, convolutional neural networks (CNNs) have achieved great success in computer vision and geoscience communities (Bolton & Zanna, 2019; Ham et al., 2019; Krizhevsky et al., 2012; Simonyan & Zisserman, 2014). Compared with other deep learning methods, CNN has the advantage to take the spatial relationship of the input image into account, making CNN especially suitable for geoscience data that are geographically distributed (Bolton & Zanna, 2019; Goodfellow et al., 2016).

Temperature and salinity are the basic physical properties of seawater, and their distributions imply the ocean dynamic processes and have essential influences on biogeochemical cycles (Stewart, 2008; Talley et al., 2011a, 2011b, 2011c). Given the scientific importance, ocean temperature and salinity structures are still poorly observed (Roemmich et al., 2009, 2019), let alone the observations at high resolution. The higher resolution (HR, hereafter) data are essential to understand oceanic processes on multiscale, for example, mesoscale and sub-mesoscales (Dickey, 2003; McWilliams, 2016; Talley et al., 2011e). Ocean temperature (or salinity) structures have been estimated in some previous studies (Akbari et al., 2017; Klemas

& Yan, 2014); however, there are still some improvements, such as the data resolution and the estimation accuracy (Akbari et al., 2017; Klemas & Yan, 2014). Most previous researches use Argo gridded data as the ground truth and thus the obtained data have a horizontal resolution of rough 1° (Lu et al., 2019; Su et al., 2015, 2018, 2019; Wu et al., 2012). The available high-resolution data are mainly from physics-based numerical models, such as assimilation and reanalysis data; however, these numerical model-based data have sources of errors themselves, such as the forcing due to the lack of observation, parameterization (e.g., eddy viscosity), and truncation error. A HR data can also be obtained by interpolation, such as traditional interpolation methods, DINEOF (Data Interpolating Empirical Orthogonal Functions) method (X. Liu & Wang, 2019), and DINCAE (Data INterpolating Convolutional Auto-Encoder) method (Barth et al., 2020); the interpolation will introduce some error per se and those methods have the drawbacks of prescribing the form of the functions. There are also some deep learning-based super-resolution methods, which have been widely used in image processing in the computer vision community (Dong et al., 2014, 2016; Ha et al., 2019; Yang et al., 2019). As a consequence, reconstructing HR temperature or salinity structures is essential.

To fully utilize the plentiful satellite products at the sea surface and alleviate the shortage in ocean interior observations, here we develop a data-driven model to estimate subsurface temperature anomaly (STA) and subsurface salinity anomaly (SSA) absolutely from satellite-observed sea surface data. We select STA (SSA), instead of the original temperature (salinity), as the target parameters to improve the accuracy because anomalies contain much smaller magnitudes and thus are harder to retrieve; and the original value can be easily obtained simply by adding back the referring mean value. The model is based on the CNN network by utilizing its ability of deep architecture, spatial relationship considering, and composing HR output. The obtained STA and SSA fields are at 26 depth levels from near-surface down to 2,000 m at a HR of $1/4^\circ$. For demonstration purposes, we selected the center of the Pacific Ocean as the research area. However, this method can also be applied to any other oceans. A series of experiments is designed in this study to comprehensively evaluate the model's performance, sensitivity, stability, and generalization. Our model successfully estimated the STA and SSA fields at a $1/4^\circ$ with remarkable accuracy; the model has stable performance during the whole research period of all seasons in 2012.

After the model is well trained, the desired STA and SSA fields are obtained simply by providing the model with the corresponding sea surface data, which are all available from satellite observations. Therefore, our method is of great simplicity and convenience by comparing it with the dynamics-based model simulation. Reconstructed data are especially valuable and useful where there are no direct measurements but satellite data are available. Here, the original temperature and salinity fields data are also obtained, and we further calculate the density fields, the dynamic height (DH or, geopotential anomaly) fields, as well as the corresponding geostrophic velocity (Talley et al., 2011e). Our study largely extends the usage of satellite observational data from two dimensions to three dimensions and exhibits the enormous potential of deep learning techniques for facilitating ocean dynamics and broader earth science studies.

2. Data and Methods

2.1. Data for Training and Evaluating the Model

Multisource satellite observations are used as inputs of the model in this study, including sea level, SST, SSS, and surface wind speed. The sea level data measured by altimeter are provided by AVISO (Hausser et al., 2021). SST data are from daily optimum interpolation sea surface temperature (OISST) data from NOAA (Banzon et al., 2020; B. Huang et al., 2021). Surface vector winds are from the Cross-Calibrated Multi-Platform (CCMP), produced based on satellite, moored buoy, and model wind data (Atlas et al., 1996, 2011). SSS are monitored from satellite by the Soil Moisture and Ocean Salinity (SMOS) mission (Boutin et al., 2018, 2020). All these sea surface data are monthly mean data (for OISST data, calculated from daily data) and at $1/4^\circ$ (or around) resolution.

The ocean subsurface temperature and salinity data from Argo observation are used here as ground truth data. We use the 1° global gridded Argo data with 27 standard levels from 0 to 2,000 m, which is also on a monthly scale and available from 2005 to the present. The data are provided by International Pacific Research Center (IPRC). Monthly temperature and salinity gridded fields from the In Situ Analysis System (ISAS) are also used for cross-validation, which is also entirely based on the Argo data and has a $1/2^\circ$

horizontal resolution (Gaillard, 2012; Kolodziejczyk et al., 2017). Argo profiles data are also used, which is from EN4 (version EN.4.2.1, the .g10. profiles), Met Office Hadley Centre observations data sets (Good et al., 2013).

The specific information and links of all above data sets are provided in the Acknowledgments section.

The zonal and meridional surface vector wind stresses are calculated from the surface vector wind data, as $\tau = \rho_a C_D |U_{10}| U_{10}$ (Smith, 1988; Trenberth et al., 1990) with ρ_a is the air density, U_{10} is the wind speed at 10 m, and C_D is the drag coefficient (Jones & Toba, 2001). The SLA, SST, and surface wind data contain the same grid information. We note that, the SSS data was interpolated to the same grids to maintain the same spatial distribution. The anomalies of surface parameters and ocean interior temperature and salinity were calculated by referring to the climatological annual-mean during 2005–2014 (for SSS, during 2010–2014). To remove the difference between the units and facilitate the training process, all input data were normalized and scaled by standard deviation before the training (Goodfellow et al., 2016).

2.2. Model Architecture

The specific structure of the CNN network in this study is shown in Table S3 in Supporting Information S1 and Figure 1. Briefly, the structure of the CNN network consists of an input layer and an output layer, as well as multiple hidden layers. The hidden layer consists of four convolutional layers, each followed by a minibatch layer and a nonlinear activation function (the rectified linear unit, ReLU), a maximum pooling layer, one dropout layer, and three fully connected layers. Each of the first two is followed by a ReLU layer and ends with one regression layer as the loss of the network. ReLU activation function is commonly used in modern neural networks, because it is both simple to implement and effective at overcoming the limitations of other previously popular activation functions, such as Sigmoid and Tanh (Goodfellow et al., 2016). ReLU remains very close to linear, as a result, it preserves many of the properties that make the models easy to optimize with gradient-based methods and the models generalize well (Goodfellow et al., 2016), and it is less susceptible to vanishing gradients that prevent deep models from being trained, although it can suffer from other problems like saturated or “dead” units. The pooling layer makes the outputs invariant and less sensitive to small translations of the input and reduces the computational amount. The fully connected layers connect all the activations in the high-level feature maps. For all convolution layers, the size of the filter is 3×3 (image height $\times$ image width) with a stride of 1. The average pooling layer has a size of 2×2 and a stride of 2. The dropout has a probability of 0.2 and the regression layer computes the half mean squared error loss (Table S3 in Supporting Information S1).

2.3. Flowchart

Before training, all the samples were randomly organized into three separate sets: training set (60%), validation set (20%), and testing set (20%). The training and validation sets are used during the training process, while the testing set is applied to evaluate the trained model's accuracy and its generalization ability (Table S1 in Supporting Information S1).

There are two steps for each experiment: model training and model evaluating. In step one, we train the CNN model by using sea surface data and labeled data (Figure 1). As a supervised regression, the loss of the network is the deviation between the estimated values and the labeled values. The network is trained by using stochastic gradient descent with momentum; a decay learning rate is employed, and a minibatch is adopted for the optimizer (Table S3 in Supporting Information S1). The training processes are monitored by plotting the root mean square error (RMSE, hereafter) of the training sets and validation sets. In step two, we evaluate the performance of the model based on the testing set, by comparing the model-estimated values with the labeled data.

After the model is well trained, the target parameters, subsurface temperature, and salinity in this study are then obtained simply by adding the new sea surface data to the obtained model. The results were evaluated using various statistical metrics (Text S1 in Supporting Information S1), including coefficient of determination (R -squared or R^2 , hereafter), correlation coefficient (R , hereafter), and RMSE. R^2 represents the proportion of the variance for a dependent variable explained by the independent variables in a regression

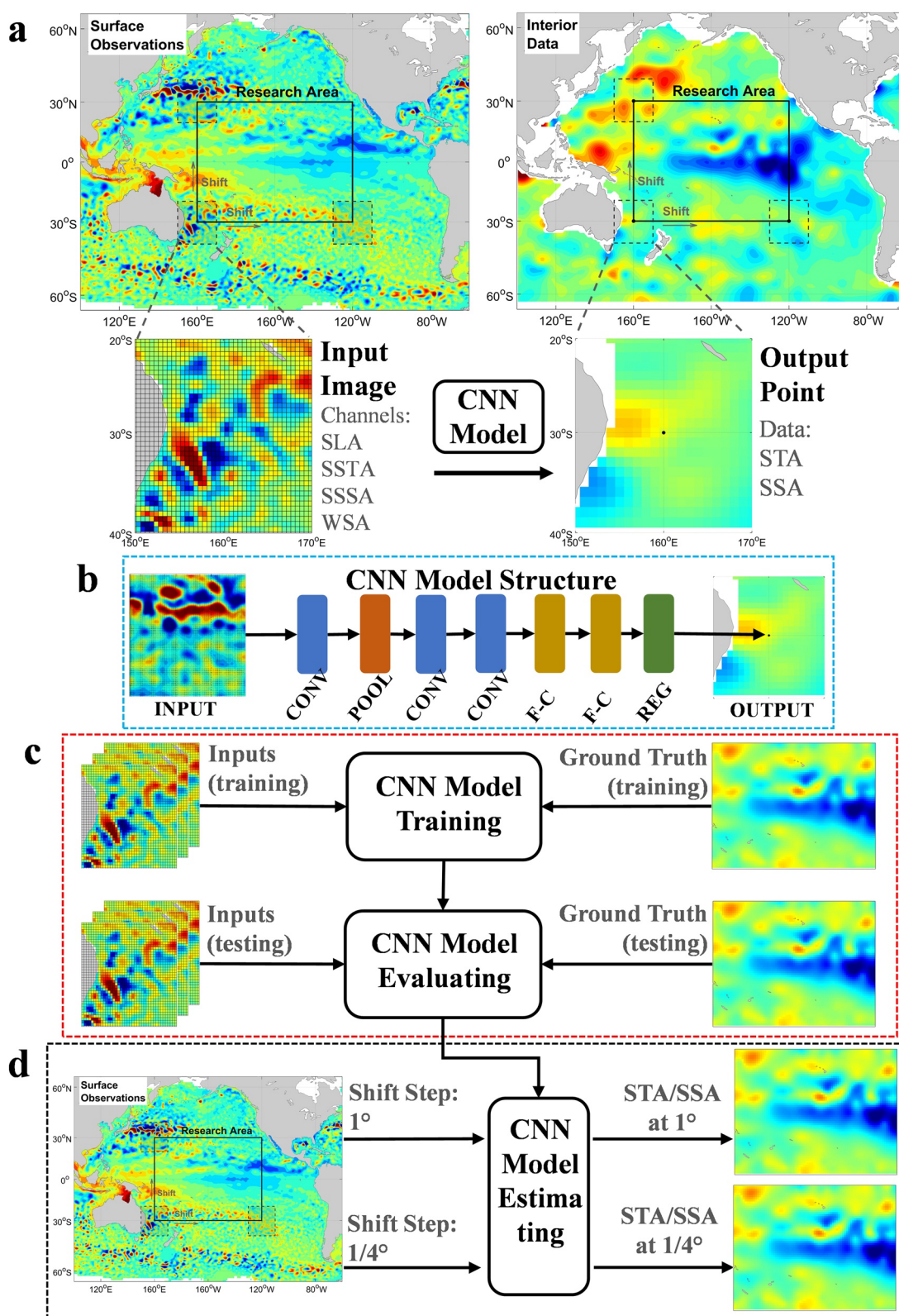


Figure 1.

model (Text S1 in Supporting Information S1); while R explains the strength of the relationship between the observed values and the predicted outcomes (Text S1 in Supporting Information S1). The RMSE describes the specific value of the errors between the predicted outcomes and the observed data.

2.4. Dynamic Height

DH, or geopotential anomaly, is created to calculate the geostrophic velocity. DH is defined as a function of density profiles:

$$DH = \int_0^p \delta(T, S, p) \cdot dp \quad (1)$$

where $\delta(T, S, p) = \alpha(T, S, p) - \alpha(0^\circ\text{C}, 35 \text{ psu}, p)$ is the specific volume anomaly. The horizontal gradients of DH represent the pressure gradient force, from which one can calculate the geostrophic velocity shear ΔV , as in equation:

$$f \times \Delta V = -g \nabla_H (DH) \quad (2)$$

with f is the Coriolis parameter vector and g is the acceleration of gravity. As a result, the geostrophic velocity at one level relative to another is then obtained from the gradient of the vertical difference of DH.

In this study, the seawater density field is calculated from the temperature and salinity fields (Fofonoff & Millard, 1983; Millero et al., 1980), based on the estimated data and the ground truth data; thus, DH field is then obtained. Here we selected 2,000 m depth as the reference level, geostrophic velocity relative to the reference level at each level is then obtained from the vertical difference of DH at each level relative to 2,000 m.

3. Results

3.1. Scheme for Higher Resolution Estimation

STA and SSA fields are estimated based on the anomalies of sea surface data including sea level, SST, SSS, zonal, and meridional wind stress (namely, SLA, SSTA, SSSA, UWSA, and VWSA). STA/SSA at each grid is estimated from the surface information within a small square box, whose horizontal center is at the target subsurface grid (Figure 1). Sea surface information within each box is the input image of the CNN model, with different variables as different channels (Figure 1, Table S1 in Supporting Information S1). Samples (input-output pairs) are made by zonally and meridionally sliding the box on the map, with a certain stride (1°), the corresponding output grid of STA/SSA also slides with the same stride (Figure 1). Images with missing value grids more than half are regarded as invalid samples and discarded; those with missing values less than half are padded with 0 before being used.

The data-driven model aims to build the relationship between the multi-source satellite data and subsurface data; more precisely, the CNN model connects the target ocean interior temperature (and salinity) to the input image of various sea surface data. The estimation procedure has two steps: model building and model estimation (Figure 1). In step one, the model is trained and evaluated based on the subsurface data from Argo, as the ground truth data of the model. The sliding stride in this step is 1° the same resolution as the ground truth data from Argo. In step two, STA and SSA are estimated by adding new sea surface data into the well-trained model. In this step, the sliding stride is $1/4^\circ$, the same as the sea surface observation data. Therefore, STA and SSA at both 1° and $1/4^\circ$ resolution are estimated (Figure 1). We treat $1/4^\circ$ as HR, relative to the 1° ground truth data that from Argo observations.

Figure 1. Scheme for the estimation. (a) Subsurface parameter at each grid is estimated from the sea surface data within a small box centered on this grid. Samples are obtained by sliding the box at the sea surface and the corresponding point at the subsurface. (b) The CNN model has 15 hidden layers (see Table S3 in Supporting Information S1). (c) The CNN model is trained and evaluated based on the data samples, surface data are from satellite observations, and subsurface ground truth data from Argo observations. (d) Estimation of data is made by providing new inputs to the well-trained model, the obtained subsurface temperature anomaly and subsurface salinity anomaly then have a horizontal resolution as high as the surface data ($1/4^\circ$). Research area in this study is in the central Pacific (160°E – 120°W and 30°S – 30°N).

We started with an assessment of the input, that is, the influence of different input images (image size, image resolution, and different variables) on the model performance. Overall, with more information involved, that is, the more grids of an input image (larger image size and higher image resolution), the better performance model achieved (Table S2 in Supporting Information S1). Moreover, as the grids of input image increased, the improvement of the performance diminished; Image size seems to have larger contributions more than image resolution does (Table S2 in Supporting Information S1). Meanwhile, as input image grids increased, the computational amount increased dramatically; a larger image size requires more area of data coverage. Furthermore, sea level plays a major role in the STA estimation, followed by SSS; however, SST and sea wind stress also have some contributions (Table S2 in Supporting Information S1). As a consequence, with the priority of accuracy and trade-off between computation amount and data availability, we set the input image with a size of $16^\circ \times 16^\circ$, at $1/2^\circ$ resolution, and involving all five sea surface variables for the estimation of STA and SSA.

3.2. STA Fields Estimation

With optimal parameters of input, we first applied our model to estimate STA at different depth levels and at different times. Figure 2 displays the performance of STA estimation at 26 depth levels from near-surface at 5 m to the deep ocean at 2,000 m, and in all months in 2012. RMSE, R^2 , and R were computed at each depth level in each month in the study area; all performance results are obtained based on the testing set. R^2 is larger than 0.98 for all research depth levels and months, the average R^2 is 0.996 for all experiments (Figure 2a). RMSE is less than 0.08°C for all depth levels and research periods, the average RMSE is 0.022°C . As the depth level increases, RMSE decreases for all the research periods (Figure 2b). Variations of the ocean are generally larger at the upper layer than at the deeper layer, as a consequence, the variations of the STA decrease from upper ocean to deeper ocean, which can be also seen from the profile of depth-wised root-mean-square (RMS) calculated from the ground truth data (Figure 2c, Figure S1 in Supporting Information S1). RMSE and the RMS calculated from the ground truth data displayed a similar series with respect to the depth, that is, RMSE was proportional to RMS at each depth level, and the ratio of RMSE to RMS is less than 6% for all depth levels (Figure 2c), indicating the estimations are accurate and very stable. For the upper ocean (50 m above), RMSE of the estimations have a slightly different depth-wised profile from the RMS (Figure 2c), this is because the estimations at these depth levels are relatively more accurate (Figure 2a). Overall, our model accurately estimated the STA fields, and the performances are very stable at different depth levels and at different times.

Figure 3 shows the distributions of the ground truth and estimated STA at different depth levels (20, 50, 100, and 150 m) in January 2012. Overall, the STA map of the estimated data is highly consistent with that of the ground truth data, they have almost the same general pattern and almost similar features and details (Figure 3). At 20 m depth, STA map is generally a La Niña-like seesaw pattern; there is a cold “tongue” at the equator area, STA is negative on the north subtropics and positive on the south subtropics, STA value is small between the cold tongue and 20°N and 20°S (Figure 3a). The estimation of STA at this depth has an R^2 value of 0.998 and RMSE as 0.0475°C , roughly 3% of the RMS value (1.52°C) of the STA in the research area. There are some slight differences between the estimated and ground truth data; for example, the model slightly underestimated the negative STA value on 165°E , at the equator (Figures 3a, 3b). At 50 m, R^2 value of 0.997 and RMSE as 0.0515°C (roughly 6% of the RMS value, 0.9°C). STA map at this depth is dominated by the cold tongue pattern, which is cooler compared with that at 20 m; while at both the negative value at north subtropics and positive value at south subtropics reduced (Figures 3c, 3d). At 100 m, the cold tongue is still the dominant pattern, however, the cold tongue has some meanders and there are also some eddy-like patches (mostly positive) scattered in the research area (Figure 3e). The STA estimation is again highly accurate, with an R^2 value is 0.997, and RMSE as 0.0421°C (Figures 3e, 3f). As the depth goes to 150 m, the “cold tongue” feature disappears on the STA map, instead, the value is main positive at the equator, with several eddy-like patches on the tropical regions; The estimation, again, highly agrees with the ground truth data (R^2 is 0.997, and RMSE as 0.0418°C), with only slight differences on some certain areas, for example, 155°W , near 30°N (Figures 3g, 3h).

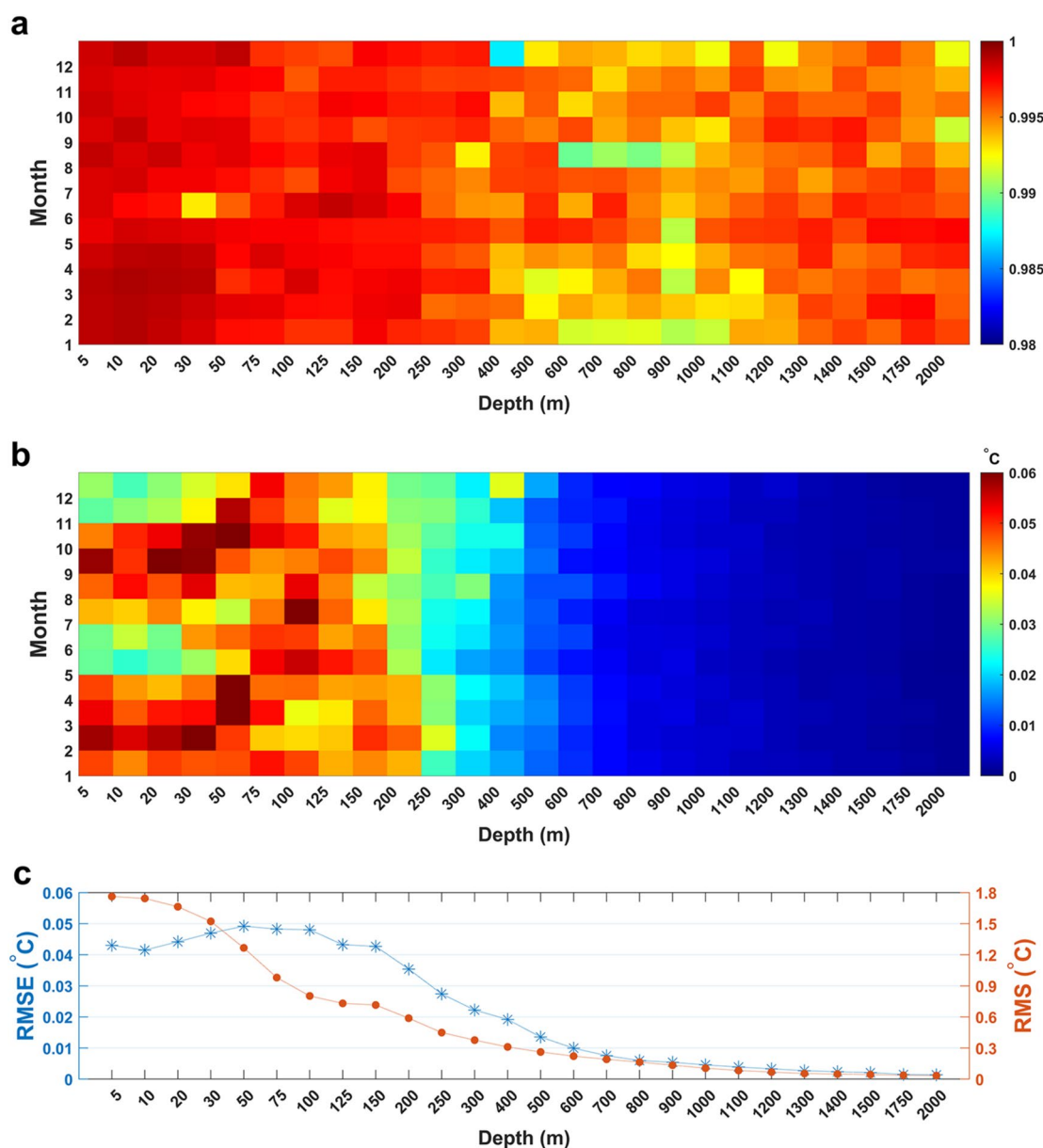


Figure 2. Model performance for subsurface temperature anomaly (STA) estimation. (a) Coefficient of determination (R^2), (b) root mean square error (RMSE) of the STA ($^{\circ}\text{C}$) estimation at each depth level and in each month of 2012. (c) Profiles of annual-averaged RMSE ($^{\circ}\text{C}$) of estimation (left, blue) and RMS ($^{\circ}\text{C}$) calculated from ground truth data (right, orange) with respect to depth level in 2012.

3.3. Higher Resolution STA Fields Estimation

As described in the model scheme, our model builds up the relationship between the image of sea surface data and the target STA, which means, the estimation is made by providing the well-trained model with the new sea surface data, precisely, each image of the sea surface data. The ground truth data from Argo is at 1° , however, the sea surface data from satellite observation are at HR ($1/4^{\circ}$). Estimation of HR STA fields are made in the same way of 1° STA estimation, the difference is that the new input image of HR estimation is obtained by shifting the image box of the surface with a stride of $1/4^{\circ}$, instead of 1° in the 1° STA estimation.

Figure 4 shows the estimated HR STA and the corresponding ground truth STA map (at 1° horizontal resolution), at 100 m depth level, in January and February 2012. The estimated results have similar overall patterns to the ground truth data, sharing the same general features, as well as the details (Figure 4).

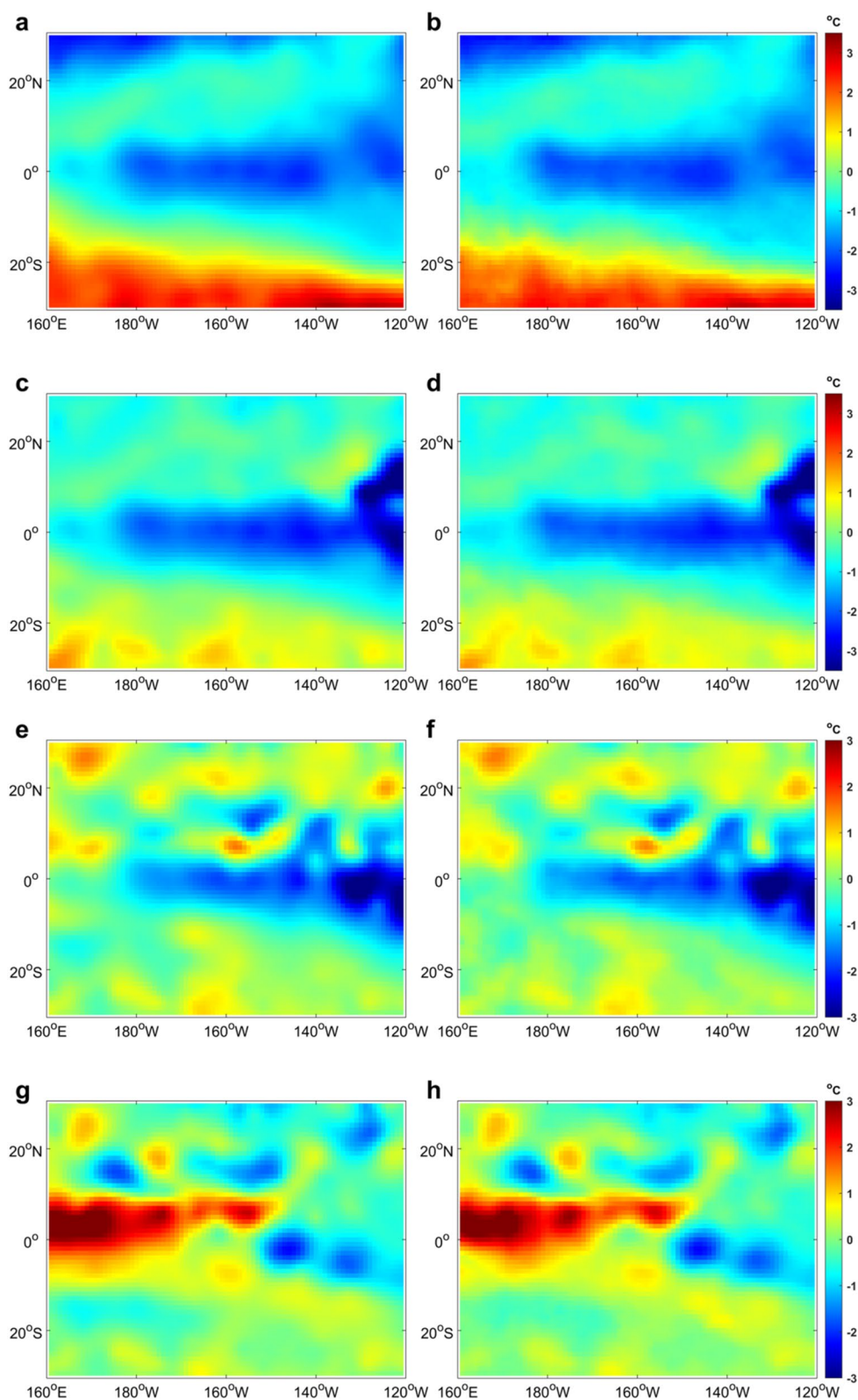


Figure 3. Distributions of ground truth and estimated subsurface temperature anomaly (STA). The distributions of STA (°C) from ground truth STA (left) and model estimated STA (right) at (a and b) 20 m, (c and d) 50 m, (e and f) 100 m, and (g and h) 150 m respectively, in January 2012 in the research area.

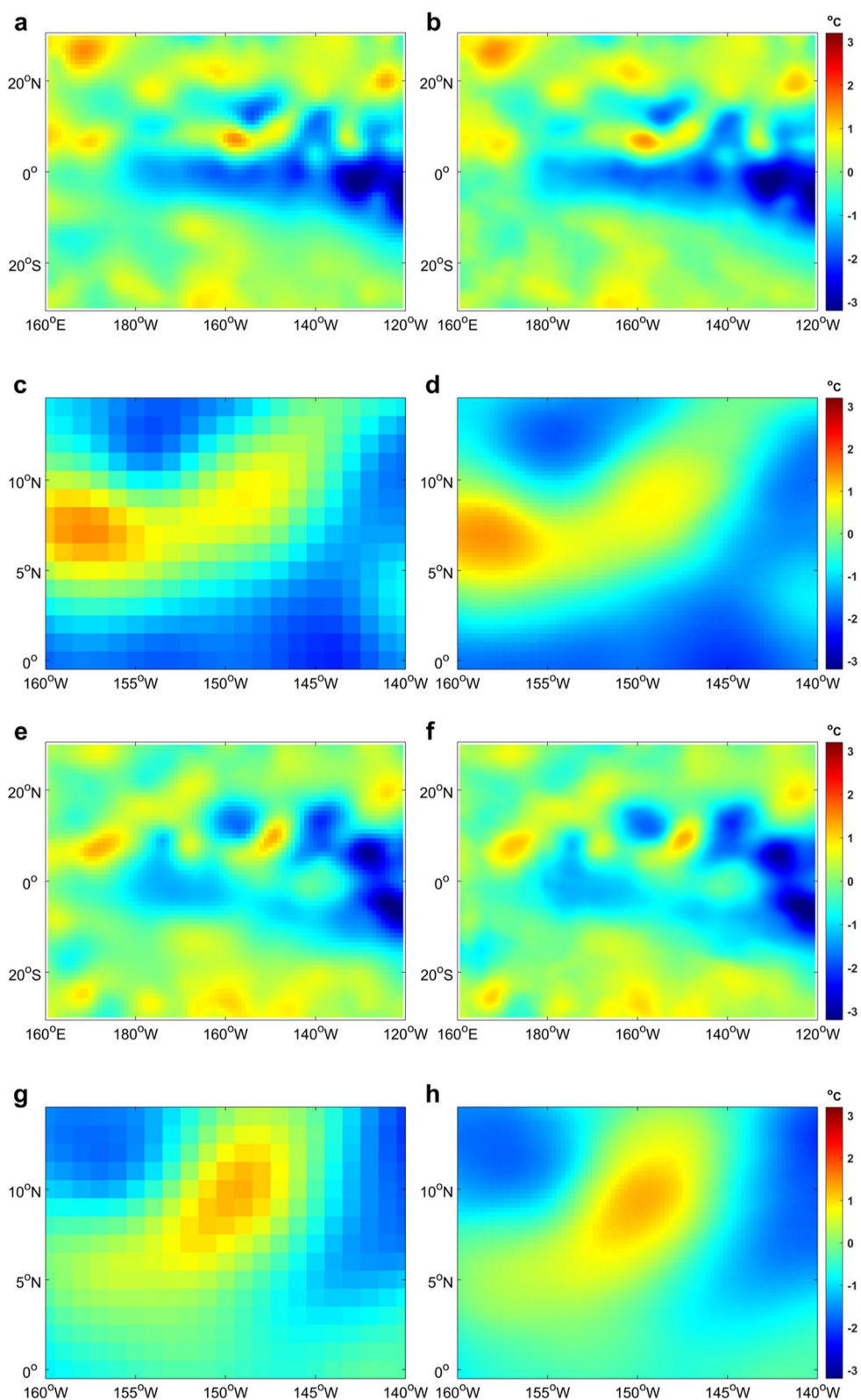


Figure 4. Distributions of estimated higher resolution (HR) subsurface temperature anomaly (STA). The distributions of (a) ground truth, and (b) HR STA ($^{\circ}\text{C}$) at 100 m in January 2012. Panels (c) and (d) are zoomed-in map of panels (a) and (b) over 0° – 15°N , 160° – 140°W ; panels (e–h) are same as panels (a–d), respectively, but in February 2012.

Patterns are smoother on the HR STA map compared with that on the low-resolution ground truth data, such differences are more evident in areas where the STA gradient is large (Figure 4). In addition, the differences are more conspicuous on the zoomed-in map at 160°–140°W, and 0°–15°N, the unsmoothed pattern on the low-resolution map is more like serration patterns, arose due to the low-resolution (Figures 4c, 4g); by contrast, the shape of the positive/negative STA values are more clear and complete on HR map (Figures 4d, 4h). The improvements in the HR are of great significance in the studies of oceanic phenomena, for example, the determination of the size and shape of oceanic eddies are more precise on the HR map, as well as the calculation of the gradients which are essential for dynamic analysis.

3.4. Higher Resolution Temperature Fields Estimation

Related to above, the temperature fields are obtained simply by adding the climatological mean values to the obtained STA fields. As a result, the estimated temperature and STA data have the same RMSE. However, since the temperature is relatively larger than STA, that is, the RMS of temperature is larger than that of STA ($\sim 5^\circ\text{C}$ to $\sim 0.6^\circ\text{C}$); therefore, R^2 (and R) calculated from the estimated and ground truth temperature are higher than R^2 (and R) between the estimated and ground truth STA (for R , 0.999–0.985). Similarly, HR temperature field is obtained by adding estimated HR STA data to the HR climatological data that is obtained based on interpolation. To further evaluate the estimation of HR temperature fields, the $1/2^\circ$ Argo gridded field data from ISAS is also compared.

Figure 5 shows the subsurface temperature map at 200 m depth in January 2012, from ground truth data, estimated data at both 1° and $1/4^\circ$ resolution, and $1/2^\circ$ ISAS Argo gridded data. The temperature map displays a seesaw pattern with high value at the western Pacific and low value at the East part, and the separation line extends southeastward and northeastward, from the near equator at 160°W to around the 15°S (15°N) at 120°W (Figure 5). The estimated temperature map is extremely consistent with the ground truth data in regards to both general pattern or main details (Figures 5a, 5b). Estimated HR temperature also displays the same pattern and feature with the ground truth map, but with the features are smoother and clearer by virtue of HR (Figure 5c). For the temperature map from ISAS Argo data, the general pattern is similar to the ground truth data that from IPRC Argo gridded data (Figure 5d); however, there are some differences in certain areas, for example, the temperature is high on 5°S, 160°–180°E, on ISAS Argo map (Figure 5d), indicating the differences in the methodology of Argo data processing. Due to its resolution of $1/2^\circ$, the features are smoother and clearer than that from IPRC Argo data and estimated 1° data, but less smooth and less clear than that of HR estimated data (Figure 5).

Comparisons between the estimated temperature profiles and the original Argo floats profiles are also made (Figure 6). The Argo profiles are on a daily time scale and the locations are relatively arbitrary, while the reconstructed data are gridded and monthly. To alleviate the effect of the time and location differences on the comparison, we randomly selected some small boxes, with a size of $5^\circ \times 5^\circ$; and the temperature profiles within the box and in a certain month are compared. Figures 6a–6c show the temperature profiles in small box 1 (170°–175°E and 20°–25°N), box 2 (175°–180°E and 2.5°–2.5°N), and box 3 (130°–125°W and 15°–20°S) in January 2012, with profiles from Argo floats (gray lines) and the constructed (red lines) respectively. There are 6, 11, and 16 Argo profiles in box 1, box 2, and box 3, respectively, and from the profiles, the mixed-layer and the thermocline can be found (Figure 6). The results show that the estimated temperature profiles are very similar to the Argo floats profiles, especially for those below 500 m. The RMSE between the averaged profiles from Argo (dash back line) and estimation (dash red line) are 0.61°C , 0.47°C , and 0.65°C in box 1, box 2, and box 3, respectively (Figure 6). The errors are partly ascribed to the differences in the time and location in the observed and reconstructed data. Overall, this model successfully and accurately reconstructed the temperature vertical profiles compared with the observed data.

3.5. Higher Resolution SSA and Salinity Fields Estimation

Like the estimation of STA, SSA is estimated as well. The same CNN model is used with the same inputs from the satellite observational data, while the output has changed from STA to SSA. The ground truth SSA data that is used to train and validate the CNN model is also from Argo gridded data. Similarly, we estimated

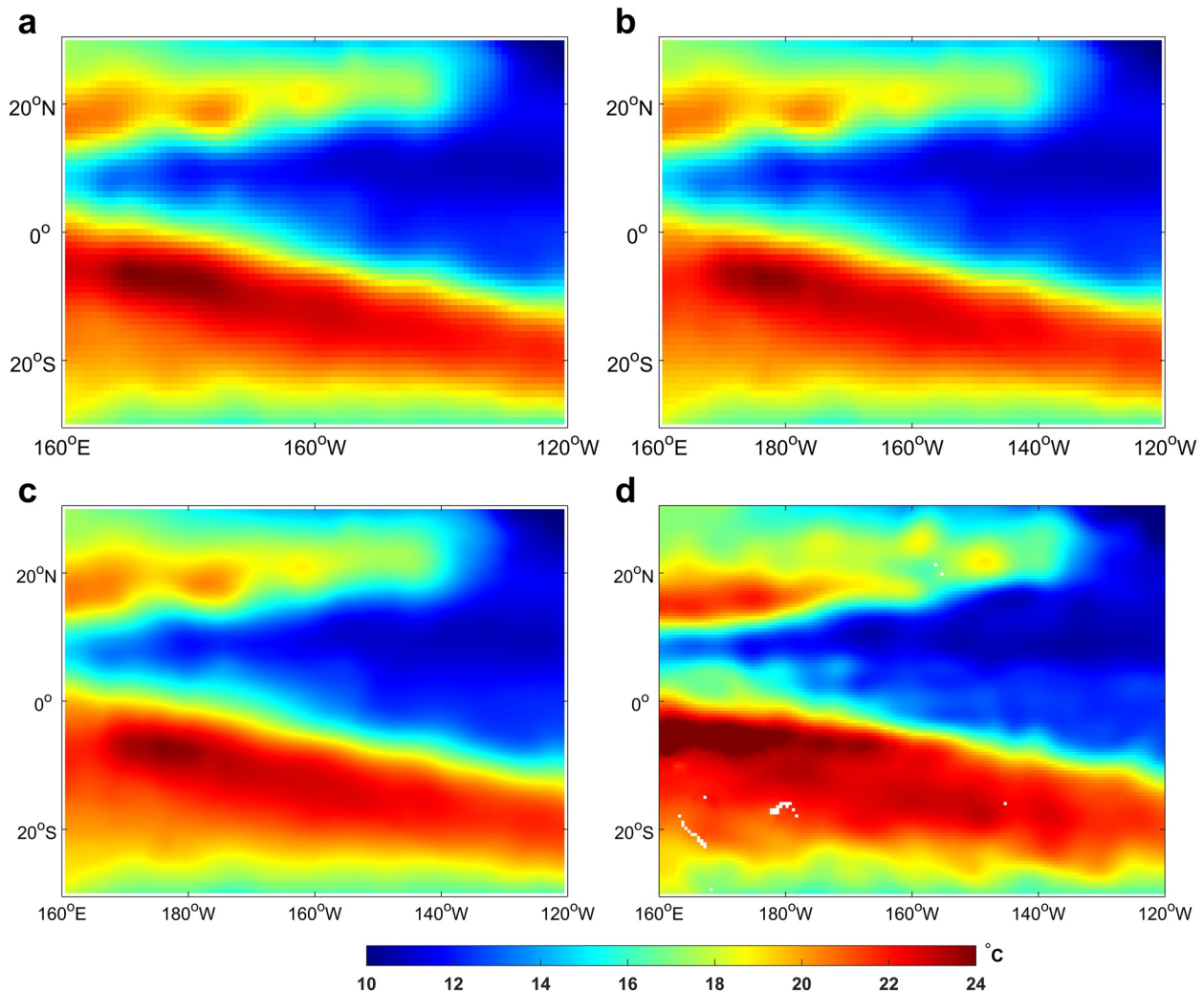


Figure 5. Distributions of estimated higher resolution (HR) temperature. The distributions of (a) ground truth, (b) 1° estimated, (c) HR (1/4°) estimated temperature (°C), and (d) temperature from ISAS Argo gridded data at 200 m depth in January 2012.

the SSA at 1° and 1/4° (HR) fields, as well as the salinity and HR salinity fields, which have 27 standard depth levels and at different times.

Figure 7 exhibits the performance of the estimation of SSA at 26 depth levels in the Pacific Ocean, from January to December in 2012. Overall, the estimation is of high accuracy. R^2 is larger than 0.97 for each depth level and each month during the research periods (Figure 7); R is larger than 0.98 for any depth level in any month during the research period (Figure S2 in Supporting Information S1). The average R^2 is 0.991 and RMSE is 0.0028 psu for all depth levels and research periods. Also, R^2 value slightly decreases with respect to the depth levels, from 0.998 at 5 m to 0.978 at 2,000 m (Figure 7a); R has a similar distribution with R^2 and is slightly larger than R^2 on each research depth level and research time (Figure S2 in Supporting Information S1). RMSE significantly decreased along with the depth levels, from 0.0074 psu near-surface to 0.0004 at 2,000 m (Figure 7b). Nonetheless, RMSE is proportional to RMS that calculated from the ground truth data with respect to the depth level, with a rate of ~5% (Figure 7c, Figure S2 in Supporting Information S1), indicating the estimations using the CNN model are very stable for different depths. The slight decrease in the depth-wised R^2 is because the value of STA decreased with respect to the depth and because very small at the deep ocean (0.0004 psu at 2,000 m depth level), such small values make the estimation harder, given that, the model can still accurately obtain the STA at the deep ocean like 2,000 m (Figure 7a).

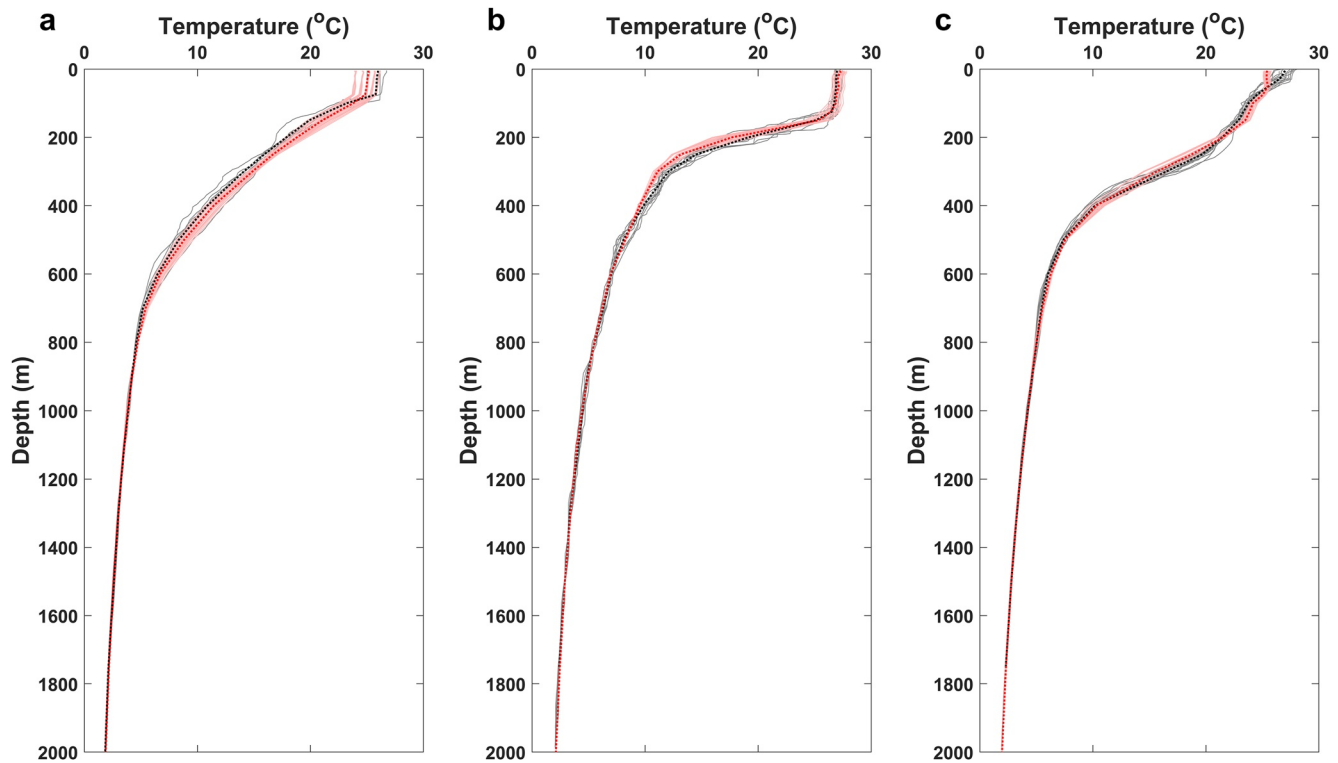


Figure 6. Profiles of the observed and reconstructed temperature. The potential temperature profiles within (a) box 1 (170°–175°E and 20°–25°N), (b) box 2 (175°–180°E and 2.5°–2.5°N), and (c) box 3 (130°–125°W and 15°–20°S) in January 2012. Temperature data are from daily Argo floats (gray lines) provided by EN4 and from the 1° estimated data (red lines). The dash lines (dash back for Argo profiles and dash red for estimated data) indicate the averaged profiles.

Figure 8 shows the distributions of ground truth and estimated SSA, as well as the estimated HR SSA, at different depth levels (50, 100, and 150 m) in January 2012. Unlike the STA map in Figure 3, there is no “cold tongue” pattern on the SSA map at the same depth and same time, instead, the SSA patterns are relatively more “disorganized” with the positive and negative value interlaces. At 50 m, the SSA is positive on the southwest part, while it is negative on the northeast part of the research area; at 100 m, it is negative near the equator and positive on other areas; at 150 m, SSA is negative around 10°S, as well as 20°–30°N, 150°–120°W, and positive on 0°–30°N, 160°E–160°W. The distributions are different from each other at different depth levels, indicating the vertical continuity is weak for these depth differences (50 m), relatively weaker than that of the SSA patterns. For the estimation, the estimated SSA captures most of the features on the map at each depth level, even at regions with extreme values and larger gradients (Figure 8). Although the estimation is of high accuracy, it still has some errors. For example, around the equator, 160°E on Figures 8a, 8b, the model slightly underestimated SSA value. The HR SSA estimation inherits the same accuracy as the estimation at normal resolution, thus, the SSA maps from them have similar patterns (Figure 8). However, since the HR estimation largely enhances the resolution, the patterns are clearer and more natural and contain more information (Figures 8c–8i). On the normal resolution map, the patterns are vague at the edge of the features or areas where the gradient is large; by contrast, features are sharper and clearer especially at the edge areas (Figure 8).

Besides, the salinity fields and HR salinity fields are also calculated based on the model results. As aforementioned relation between estimated temperature and STA, the estimated salinity and SSA have a similar relation. First, the estimated salinity data inherits the same error as the SSA and thus they have the same RMSE; however, since the RMS of the salinity is larger than that of SSA, the estimated salinity data have a higher R^2 and R values than those of the estimated SSA. Figure 9 shows the salinity distribution from the ground truth data from Argo gridded data, estimated salinity data, HR salinity data, and Argo gridded data from ISAS at 1/2° resolution at 200 m in January 2012. Unlike the relatively disorganized SSA map

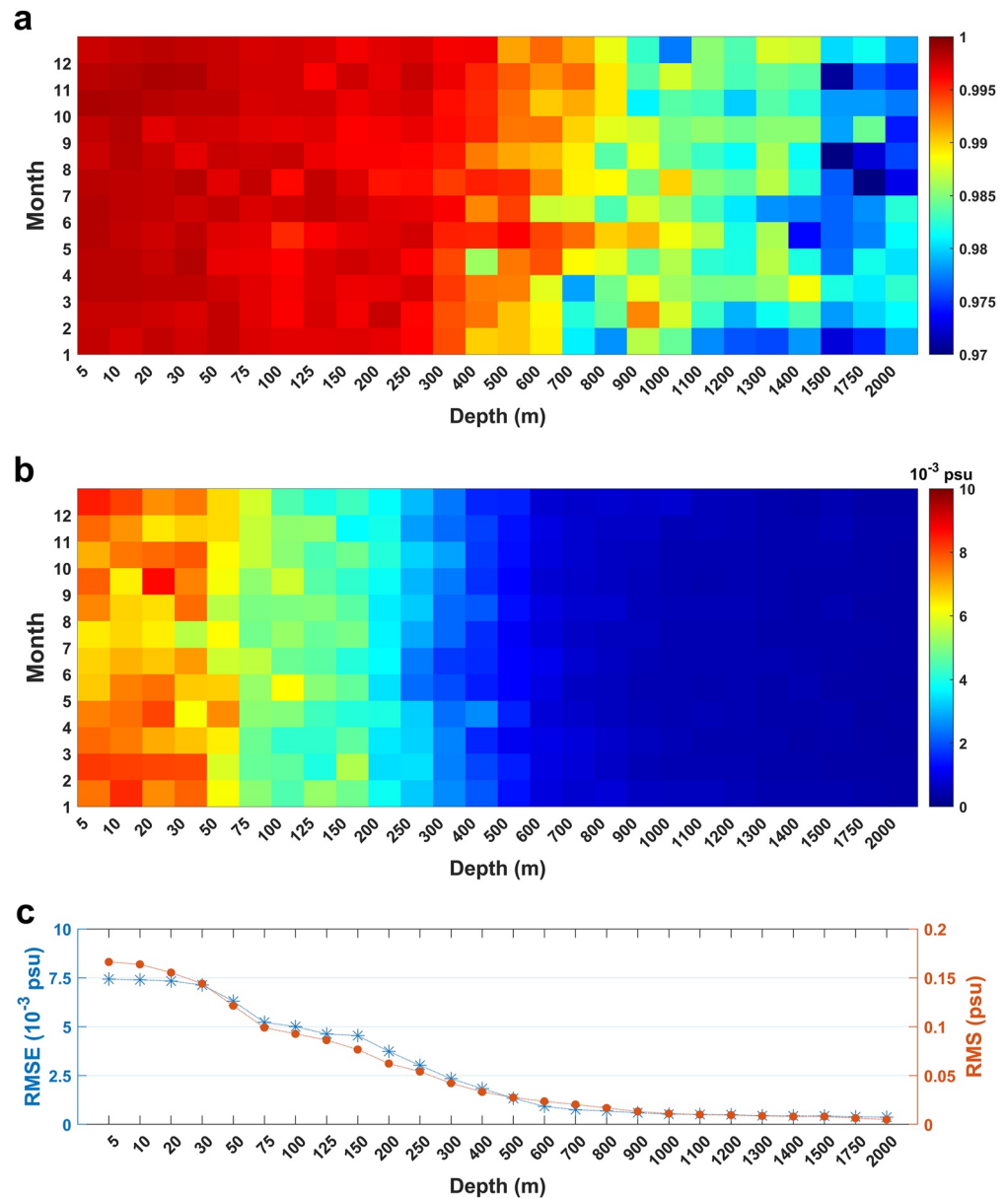


Figure 7. Model performance for subsurface salinity anomaly (SSA) estimation. (a) Coefficient of determination (R^2), (b) root mean square error (RMSE) of the SSA (psu) estimation of each depth level and in each month of 2012. (c) Profiles of RMSE (psu) of estimation (left, blue) and RMS (psu) calculated from ground truth data (right, orange) with respect to depth levels averaged over all months, 2012.

(Figure 9), the salinity displays a seesaw pattern like the temperature map in Figure 5, with high salinity at the south and low value at the north.

3.6. Dynamic Height and Geostrophic Flow Fields

DH or geopotential anomaly is calculated based on the density fields and it contains all the horizontal and vertical variations of density, while the density fields are obtained from the temperature and salinity fields. As a result, the horizontal gradients of the DH represent the horizontal pressure gradient force. Figure 10 presents the DH map at different depth levels (100, 200, and 500 m) in January 2012, derived from the temperature and salinity fields from the ground truth and the estimated data. The DH at 100 m has a two-peak distribution meridionally with a high value at 10°N and 10°S , and higher at the west part than the east part

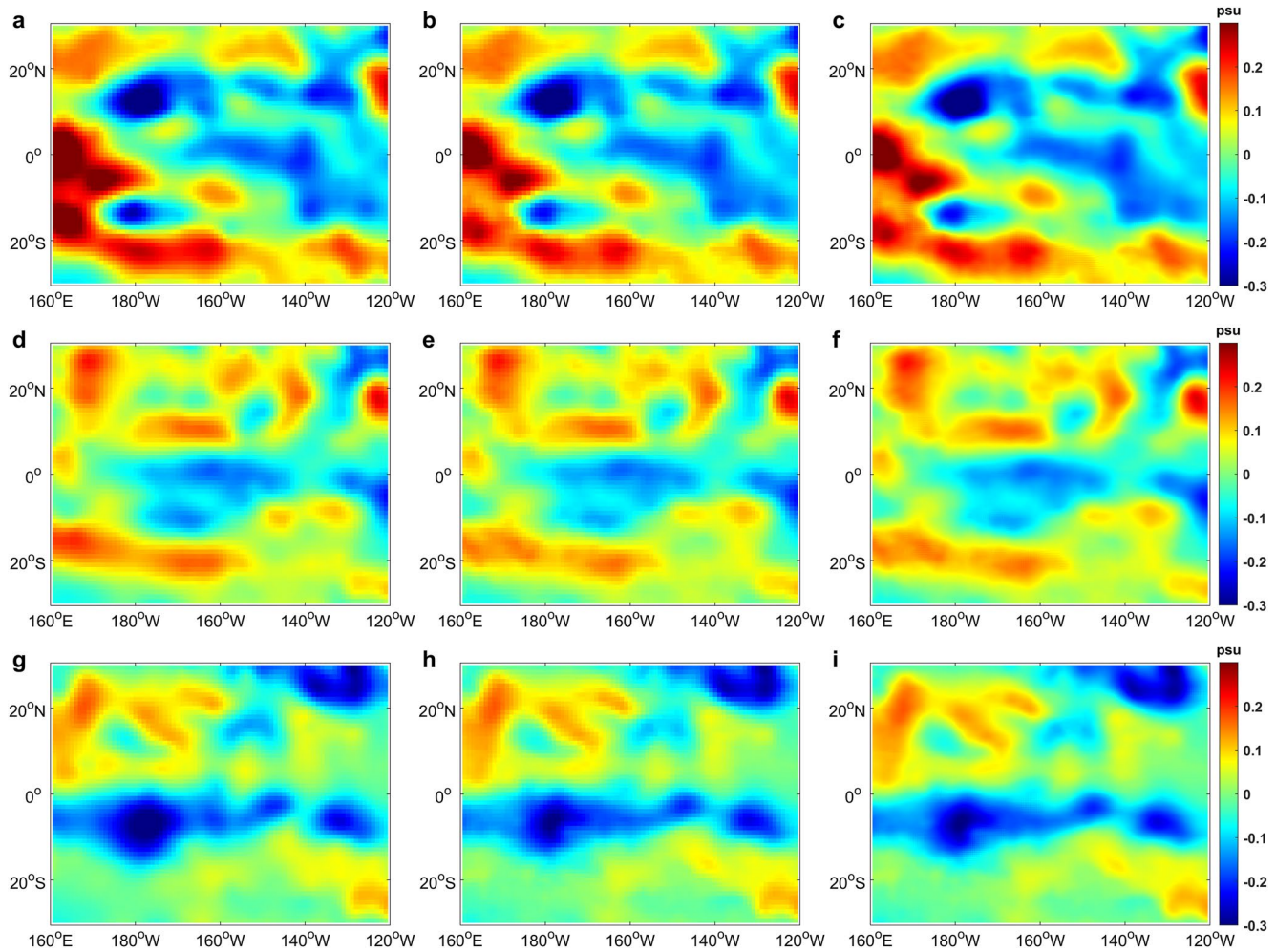


Figure 8. Distributions of ground truth and estimated subsurface salinity anomaly (SSA). The distributions of SSA (psu) from (a) ground truth data, (b) estimated data, and (c) higher resolution estimated data at 50 m depth in January 2012. Panels (d–f) are same as panels (a–c), but at 100 m and panels (g–i) are same as panels (a–c), but at 150 m.

(Figure 10a). This is a combined consequence of the distributions of both temperature and salinity in the upper 100 m (Figures 4, 8). A high temperature (or low salinity) leads to a low density and then a high DH, and vice versa. DH at 200 m has a similar pattern to that at 100 m in general, indicating the vertical continuity (Figure 10d); however, they also have significant differences in detail, as well as the range of value, indicating the uneven distribution of temperature and salinity between these two depths (Figures S3a–S3d, S4a–S4d in Supporting Information S1). At 500 m, the high DH value at South Hemisphere extends to the east part (Figure 10g); around the equator, the zonal gradient of the DH is not increasing from 200 to 500 m, suggesting the relative low gradient of temperature and salinity during these two depth levels (Figures S3e–S3h, S4e–S4h in Supporting Information S1). Besides, estimation at both normal resolution and HR are also highly consistent with the ground truth data (Figure 10).

DH difference between one level with another level is also calculated. The DH difference is the streamfunction of the geostrophic velocity at one level relative to another. We selected 2,000 m depth as the reference level in this study, which is the deepest level of core Argo measurement. Figure 11 displays the DH difference, relative to 2,000 m, at 100, 200, and 500 m depths respectively in January 2012. The distribution of the DH difference from the estimated temperature and salinity structure accurately agrees with that from the ground truth data. Figure 12 exhibited the distribution of the geostrophic currents calculated from both the Argo gridded data and the reconstructed data. The results indicate our estimation can also be applied in the estimation of the geostrophic velocity fields. Compared to the ground truth data, the HR estimation

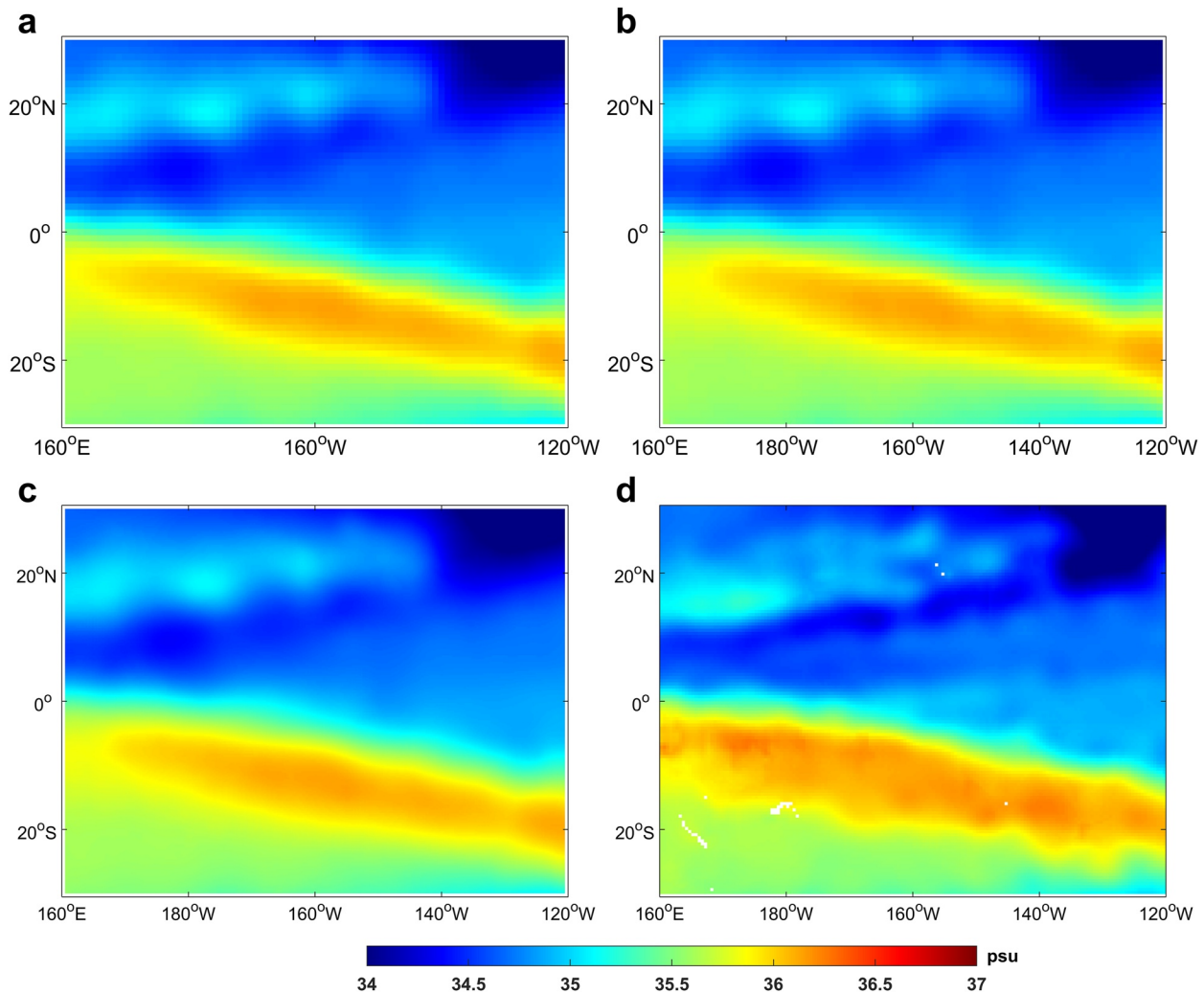


Figure 9. Distributions of estimated higher resolution (HR) salinity. The distributions of (a) ground truth, (b) 1° estimated, (c) HR (1/4°) estimated salinity (psu), and (d) salinity from ISAS Argo gridded data at 200 m depth in January 2012.

provides more information and details on the map and thus the features are clearer, especially at the feature edges or areas where gradients are large, such as the highest value over 160°–170°E, 20° and 30°N on Figures 11a–11c, and 170°–160°W, 10°–20°N on Figures 11g–11i. These improvements are critical cause the gradients are associated with the velocity calculations.

4. Discussion and Conclusion

Accurately estimating ocean interior temperature and salinity structures is of great importance for ocean science and climate studies, for the direct observations are difficult and costly and thus still very sparse. The ocean interior parameters are dynamically connected with the sea surface data; the sea surface data are plentifully collected by satellite measurements. Therefore, the idea came to researchers that retrieving subsurface information from sea surface data, for making up for the insufficiency of ocean interior data and fully utilizing the increasing volumes of satellite data. However, retrieving subsurface information from the sea surface is challenging because the direct physical relationship between sea surface data and subsurface information is still not fully understood. Nonetheless, the surface-subsurface relationship is also engraved on the data. The relationship thus can be learned from the data by machine learning methods. Here, we design a data-driven deep learning model that builds the relationship between subsurface parameters (temperature and salinity) and sea surface data from satellite observations (sea level, SST, SSS, and

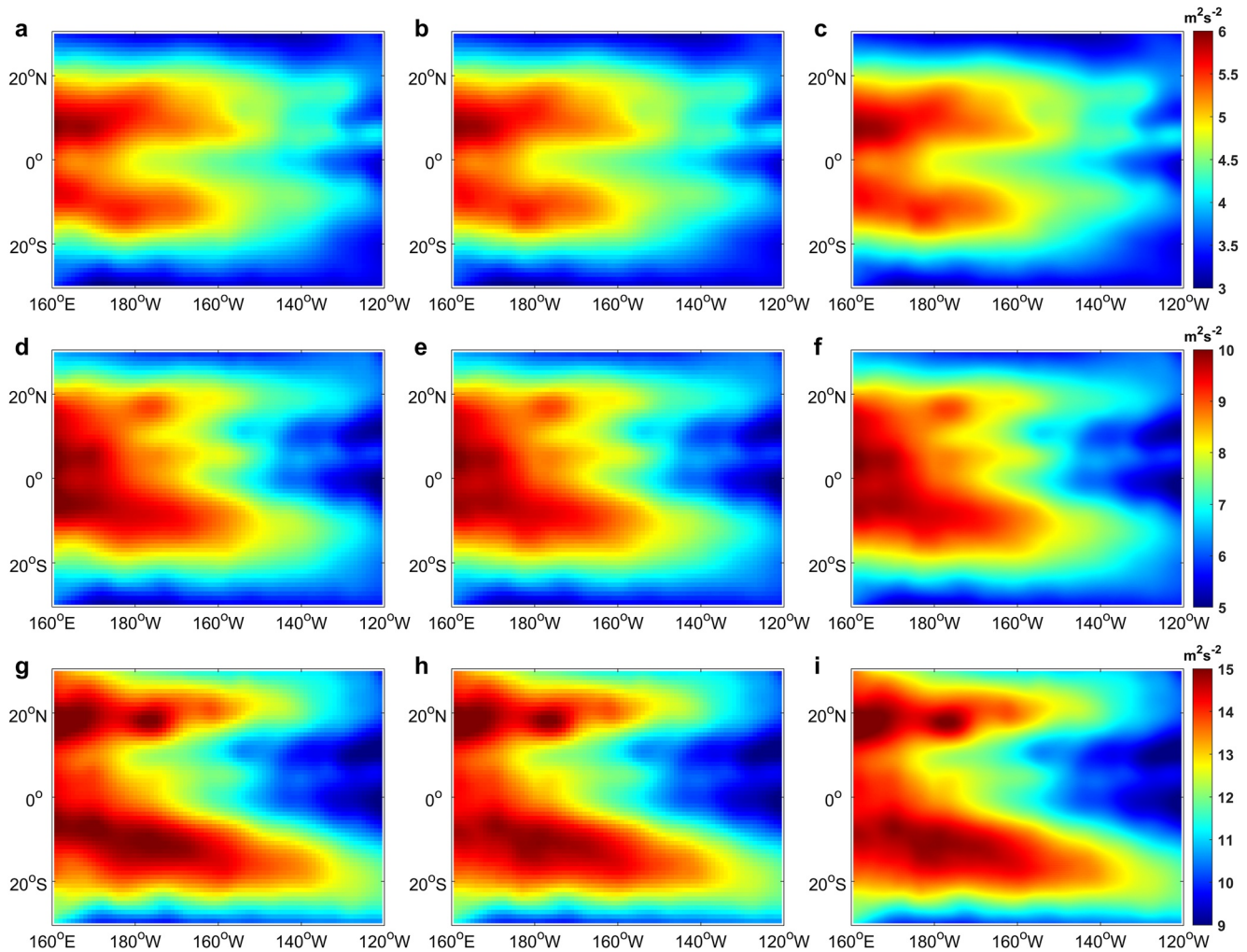


Figure 10. Distributions of dynamic height (DH). DH (m^2s^{-2}) map at 100 m depth in January 2012, derived from (a) ground truth, (b) estimated, and (c) higher resolution estimated temperature and salinity fields. Panels (d–f) are same as panels (a–c), but at 200 m and panels (g–i) are same as panels (a–c), but at 500 m.

surface wind); subsurface data are then obtained simply by combining the well-trained model with the satellite-observed sea surface data. In this study, the estimated 3D temperature and salinity fields are from near-surface to 2,000 m vertically and monthly at different times. Moreover, we develop a new scheme that successfully improved field data estimation from 1° to $1/4^\circ$ horizontal resolution, a HR level compared to the 1° Argo gridded data.

Although estimating subsurface and deep ocean parameters from sea surface data has been explored in many previous studies through both statistical and dynamical approaches (Lu et al., 2019; Su et al., 2015, 2019; Wu et al., 2012), our method has significant improvement in the data resolution and accuracy. To build up and validate any models, ground truth subsurface data are always needed, thus, the estimated data in most previous researches are also at the same resolution as the Argo data, rough 1° (Han et al., 2019; Lu et al., 2019; Su et al., 2015, 2019; Wu et al., 2012). To overcome the limitation of the ground truth data resolution, we design a new scheme that subsurface data at each grid is estimated from one small image (Figure 1); the HR subsurface data field is obtained by sliding the image with a stride of $1/4^\circ$ at the sea surface (Figure 1). Furthermore, our model substantially improved the accuracy of estimation compared to many previous studies (Han et al., 2019; Lu et al., 2019; Su et al., 2015, 2019; Wu et al., 2012). Compared with other machine learning methods, CNN model has conspicuous advantages in processing image or image-like data due to the ability to take the spatial relation into account; also, it has a larger capacity that can be tackling very complex relations compared with traditional machine learning methods (Goodfellow et al., 2016; Krizhevsky

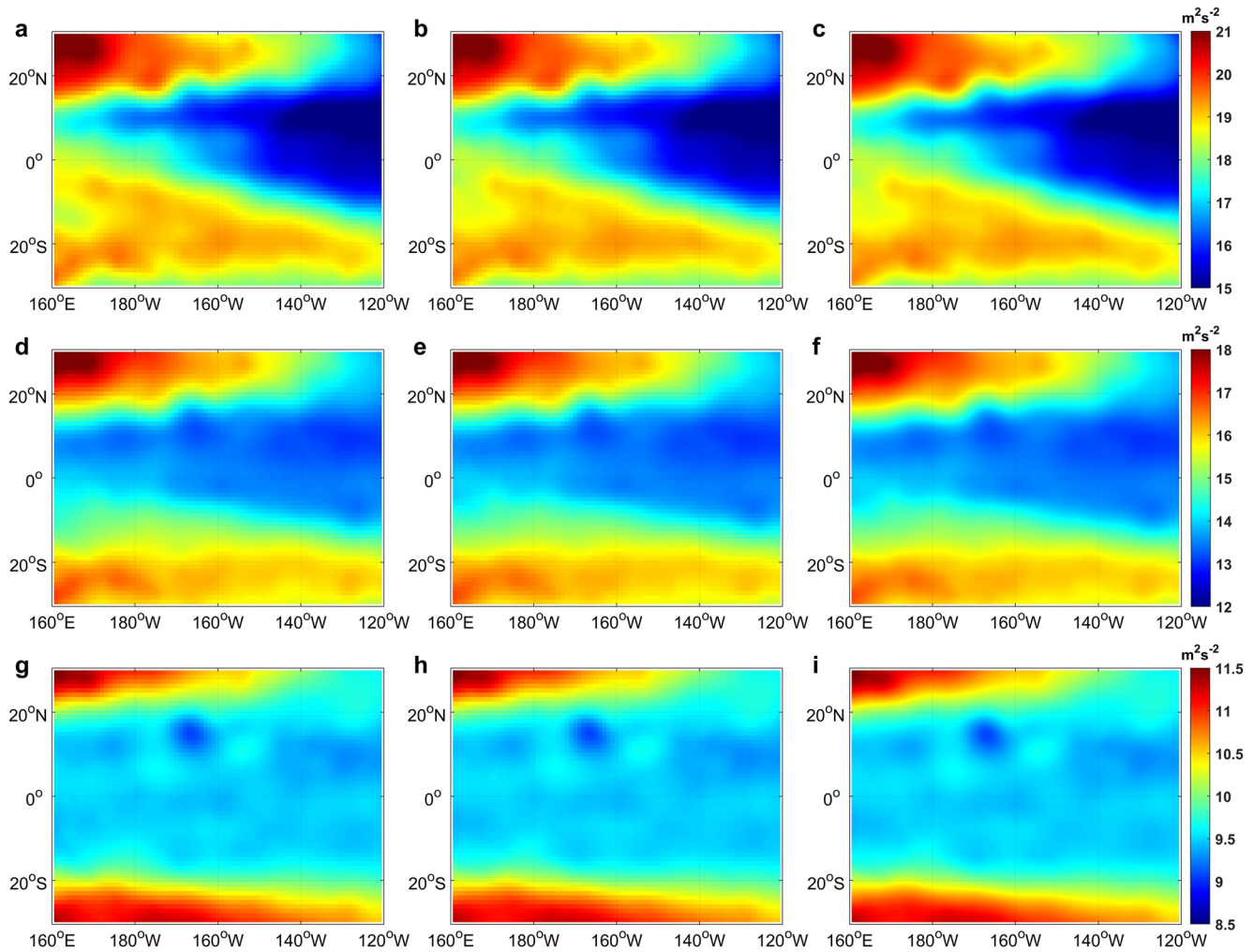


Figure 11. Distributions of dynamic height (DH) differences. DH difference (m^2s^{-2}) map at 100 m depth in January 2012, that derived from (a) ground truth data, (b) estimated data, and (c) higher resolution estimated data. Panels (d–f) are same as panels (a–c), but at 200 m and panels (g–i) are same as panels (a–c), but at 500 m. DH difference is calculated relative to 2,000 m, thus, the DH difference can serve as the stream function of the geostrophic flow relative to 2,000 m.

et al., 2012). The accuracy is improved also because we use one image to estimate each subsurface data, instead of estimating subsurface data within an image.

Our method has wide usage. First, retrieving the subsurface temperature and salinity data where there are no direct measurements but satellite data are available. For example, before Argo era, observations of subsurface temperature and salinity are very sparse while satellite-observed sea surface data are abundant. Even during Argo era, the observations are still sparse, especially in certain areas like high latitude in both hemispheres, compared with the copious sea surface data. Second, our model estimates both temperature and salinity fields, thus, as aforementioned results, the density fields and the dynamic fields can be also derived, as well as the geostrophic velocity fields. Third, the HR 3D field data have broadened the spatial scale to a mesoscale level, providing more data for studies on this scale and the processes interactions between different scales.

Remarkable performances for estimation subsurface data on a monthly time-scale are found in our model, however, data on different time-scales are also of great importance because oceanic processes are in a wide range, such as daily time-scale data are needed for studies synoptic-scale oceanic processes. Besides, further studies on the predictions of the ocean data on different temporal and spatial scales are needed to fully make use of the deep learning models.

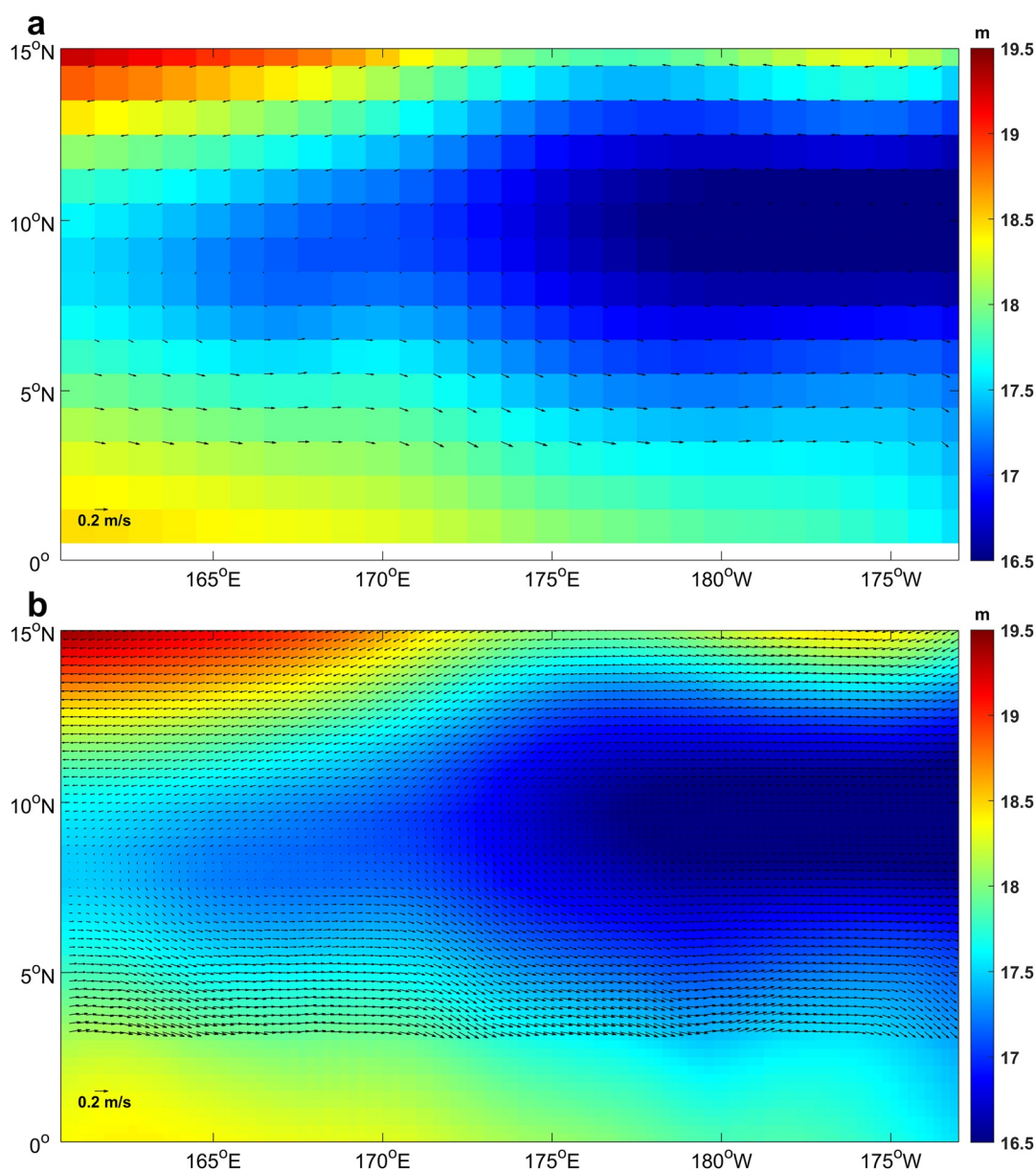


Figure 12. Distributions of geostrophic currents. Dynamic height difference (m^2s^{-2}) map (color) and the corresponding geostrophic currents (arrow) at 100 m depth in January 2012, (a) from 1° Argo gridded data, (b) from the $1/4^\circ$ estimated data.

This study extends the usage of the satellite data, enriches the ocean data for both dimension and resolution, and provides some new insights into the ocean dynamics from a data-driven aspect. Multidisciplinary researches have enormous potential, here we provide a good example between remote sensing, ocean science, and deep learning.

Data Availability Statement

The authors thank OISST data from the NOAA for sea surface temperature data (<https://www.ncei.noaa.gov/data/sea-surface-temperature-optimum-interpolation/v2.1/access/avhrr/>), CCMP (version 2) for SSW vector analysis product (<https://data.remss.com/ccmp/v02.0/>), SMOS (data set: CEC-Locean L3 Debiased v2) for SSS data (<ftp://ext-catds-cecos-locean:catds2010@ftp.ifremer.fr/>), Archiving, Validation, and

Interpretation of Satellite Oceanographic data (AVISO) for sea level data, available at Copernicus Europe's eyes on Earth (https://resources.marine.copernicus.eu/?option=com_csw&view=details&product_id=SEALEVEL_GLO_PHY_L4_REP_OBSERVATIONS_008_047), and EN4 (data set: version EN4.2.1, .g10.) for Argo profiles data (<https://www.metoffice.gov.uk/hadobs/en4/download-en4-2-1.html>). The authors also thank the IPRC (http://apdrc.soest.hawaii.edu/projects/Argo/data/gridded/On_standard_levels/index-1.html) and the ISAS15 (<https://www.seanoe.org/data/00412/52367/>) for Argo data of monthly temperature and salinity gridded fields. Argo data were collected and made freely available by the International Argo Program and the national programs that contribute to it, the Argo Program is part of the Global Ocean Observing System.

Acknowledgments

This research was supported by The National Key R&D Program of China (2019YFA0606702) and the National Natural Science Foundation of China (91858202, 41630963, and 41776003). XHY is supported by NSF (#IIS-2123264) and NASA (#80NSSC20M0220).

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